Mouse Liver Digestion Buffer with Collagenase IV

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Protocol ID: BUF-LIV-DIG-COL-IV
Version: v1.0
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Purpose

This buffer is used for enzymatic digestion of mouse liver tissue to generate viable single-cell suspensions. It contains Collagenase IV and DNase I at working concentrations suitable for dissociating liver parenchyma while preserving lymphocyte viability.

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Linked protocols

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Warning: package 'ggplot2' was built under R version 4.5.2
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Warning: package 'tibble' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

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Warning: package 'lubridate' was built under R version 4.5.2

None
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Working buffer composition

Component	Final concentration	Notes
Collagenase IV	0.5 mg/mL (example)	Adjust per validated stock; see below
DNase I	25–50 µg/mL	Reduces clumping; protects viability
PBS (1×), sterile	—	Solvent

> **Note**
>
> The final working concentrations here follow the proportions described in the original SOP (100 µL Collagenase IV into 10 mL PBS + 50 µL DNase I). Adjust exact activity units according to vendor.

Preparation

Stock solutions

Reagent	Typical stock concentration	Storage
Collagenase IV	100 mg/mL in PBS or HBSS	Aliquoted at −20 °C
DNase I	10 mg/mL in buffer (manufacturer)	Aliquoted at −20 °C

Preparation of 10 mL working buffer (2×)

1. Thaw aliquots of Collagenase IV and DNase I on ice.
2. In a sterile tube, add:
   - **100 µL** Collagenase IV stock
   - **50 µL** DNase I stock
3. Add **PBS up to 10 mL total volume**.
4. Mix gently by inversion; **do not vortex**.

This 2× buffer is diluted 1:1 with PBS at the time of digestion (2 mL 2× buffer + 2 mL PBS per liver sample).

Storage and stability

- Store freshly prepared digestion buffer **on ice** during use.
- Do **not** freeze after mixing with enzymes.
- Discard after same-day use.
- Collagenase IV and DNase I **stock solutions**:
  - Store at **−20 °C, aliquoted**, avoid repeated freeze–thaw cycles.

Reagent details

Reagent	Supplier	Cat. #	Notes
Collagenase IV	Various	—	Liver digestion enzyme
DNase I	Various	—	Prevents clumping
PBS, sterile	Various	—	Solvent

Safety

- Handle enzymes with gloves and eye protection to avoid skin or mucosal irritation.
- Dispose of enzyme-containing waste as biological/chemical waste according to institutional guidelines.
- Use sterile technique inside a biosafety cabinet.

Version history

Version	Date	Author	Changes
v1.0	2025-11-21	Dillon Corvino	Initial Quarto buffer document.